

Multiple choice questions (MCQs):

Choose the correct answer (20 MCQs): Total 10 marks (0.5 mark for each question)

1. A 47-year-old male patient presented with dyspnea, fever, dry cough, and pleuritic chest pain. The condition was associated with hyponatremia. The most probably causative organism is:
 - a. Legionella pneumonia
 - b. Streptococcal pneumoniae
 - c. Staph aureus
 - d. mycoplasma
2. Which of the following is a complication to chronic bronchitis:
 - a. Pneumothorax
 - b. Cystic fibrosis
 - c. Lung cancer
 - d. Cor pulmonale
3. Patients with hypertrophic obstructive cardiomyopathy (HOCM) may present with:
 - a. Stroke
 - b. Hemoptysis
 - c. Sudden death
 - d. Supraventricular tachycardia
4. Female patient, 65 years old, known to be hypertensive (uncontrolled due to irregular treatment), presented at an emergency room (ER) with acute dyspnea and orthopnea. On examination, the patient was cyanosed and his blood pressure was 200/100 mmHg. By auscultation, bilateral fine basal crepitation was heard on both lungs. The most likely diagnosis is:
 - a. Acute right side heart failure
 - b. Acute left side heart failure
 - c. Bronchial asthma
 - d. Myocarditis
5. Arterial blood gases (ABG) in ARDS shows:
 - a. Hypoxemia with hypercapnia

- b. Hypoxemia with hypocapnia
 - c. Hypercapnia with no hypoxemia
 - d. No hypoxemia and no hypocapnia
6. Which is the most common cause for predominant neutrophils in pleural effusion?
- a. Malignancy
 - b. Parapneumonic
 - c. Heart failure
 - d. Rheumatoid arthritis
7. Which of the following is goal of oxygen therapy?
- a. Correcting hypoxemia
 - b. Correcting underlying disorders
 - c. Correcting anemia
 - d. Correcting cardiac output
8. Horner's syndrome is a complication of which of the following?
- a. Bronchial carcinoma
 - b. Right apical carcinoma
 - c. Apical lung abscess
 - d. Right lower lobe carcinoma
9. Which of the following is the cause of using long acting beta 2 agonist with inhaled corticosteroids?
- a. Delay up regulation of receptors
 - b. Enhance down regulation of receptors
 - c. Prevent occurrence of tolerance
 - d. Decrease the hypokalemic effects
10. Which of the following chest X-ray findings is indicative of a poor prognosis in a case of interstitial lung disease?
- a. Bibasilar reticular pattern
 - b. Honeycombing
 - c. Nodular pattern of alveolar filling

3. Lung abscess () **C**

4. Bronchogenic carcinoma () **F**

• Options:

- (A) Chest X-ray
- (B) Sputum zeil Nelson test
- (C) High-resolution CT chest (HRCT)
- (D) Pulmonary function test (PFT)
- (E) CT pulmonary angography (CTPA)
- (F) Lung biopsy

Q2: For each clinical scenario, you should choose the most appropriate option from the option list, and you can use the lettered option once, more than once, or not at all (i.e. an option can be correct for more than one clinical scenario and another option can be incorrect for all scenarios). Total one mark (0.5 mark for each scenario)

I. A 65-year-old lady presented with shortness of breath on exertion, fatigue, and palpitations. On examination, her jugular venous pressure was elevated, and a pansystolic, high-pitched "whistling" murmur was heard. The murmur radiated to the left axilla. () **A**

II. ECG identified by the PR interval tends to become longer with every succeeding ECG complex until there is a P wave not followed by a QRS. () **F**

• Select the most likely diagnosis from the following list:

- a) Mitral regurgitation.
- b) Mitral stenosis.
- c) Aortic regurgitation.
- d) Hypertrophic cardiomyopathy.
- e) Second-Degree Atrioventricular Block, Type II.
- f) Second-Degree Atrioventricular Block, Type I.
- g) First-Degree Atrioventricular Block.

Good Luck

● Short Essay Question (SAQ):

Write in the following (3 SAQs): Total 3 marks (1 mark for each question)

Q1: Define acute massive pulmonary embolism?

Q2: Enumerate four causes of pneumothorax?

1. _____
2. _____
3. _____
4. _____

Q3: List the major and minor Jones criteria for diagnosis of acute rheumatic fever?

● Extended Matched Question (EMQ):

Total 2 marks (One mark for each question)

Q1: Each group of extended matching questions (EMQs) consists of a list of problems (diseases) followed by a lettered set of options (diagnostic tests). For each numbered disease, select the lettered diagnostic test that most closely diagnoses the disease. You can use the letter option once or not at all. Total one mark (0.25 mark for each problem)

• Problems:

1. Pulmonary tuberculosis
2. Bronchiectasis

()

()

b

A

- b. Chest x ray
- c. Arterial blood gases
- d. ECG

16. The apex is formed mainly by?

- a. Left atrium.
- b. Right atrium.
- c. Left ventricle.
- d. Right ventricle.

17. The murmur associated with mitral valve stenosis is?

- a. Mid diastolic click and late diastolic murmur heard at apex.
- b. Mid Diastolic rumbling Murmur.
- c. Holosystolic murmur.
- d. Ejection systolic Murmur.

18. What are the components of tetralogy of Fallot?

- a. Infundibular stenosis, ventricular septal defect, overriding aorta, right ventricular hypertrophy.
- b. Atrial septal defect, cleft tricuspid valve, patent ductus arteriosus, aortic stenosis.
- c. Atrial septal defect, pulmonary stenosis, ventricular septal defect, aortic regurgitation.
- d. Left ventricular hypertrophy, infundibular stenosis, overriding aorta, aortic stenosis.

19. Which of the following Duke criteria is a major criterion for infective endocarditis

- a. Immunological findings such as glomerulonephritis.
- b. Predisposing cardiac conditions or intravenous drug abuse.
- c. Chronic bronchitis
- d. Blood culture positive for typical infective endocarditis organisms.

20. What is the cardiac ejection fraction?

- a. It is the percentage of blood leaving the heart in each cardiac cycle.
- b. It is the percentage ratio of cardiac work to total energy expenditure.
- a. It is the cardiac output per square meter body surface area.
- b. It is a premature contraction of the heart occurring during diastole.

d. Mixed pattern of alveolar filling

1. Rigid bronchoscopy is preferentially indicated for?

- a. Persistent pneumonia
- b. Extracting foreign bodies
- c. Atelectasis
- d. Interstitial disease

12. Which of the following is the most common presenting feature of cystic fibrosis?

- a. Failure to thrive
- b. Persistent respiratory symptoms
- c. Meconium ileus
- d. Hepatobiliary disease

13. What would be an appropriate first-line treatment for the management of cyanosis as a result of heart disease?

- a. Indomethacin administration
- b. Prostaglandin E1 infusion
- c. Nitric oxide
- d. Surfactant

14. A previously healthy 4-year-old boy is noted to have a respiratory rate of 40 breaths per min, a heart rate of 140 beats per min, two weeks after having a nonspecific upper respiratory tract infection. His heart rate is occasionally irregular and a gallop rhythm is noted. A 1/6 blowing systolic murmur is heard at the apex. He has moderate hepatomegaly. Which of the following is the MOST likely diagnosis?

- a. Acute rheumatic fever
- b. Infective endocarditis
- c. Viral myocarditis
- d. Pericarditis

15. A 50-year-old male who is known to be diabetic, hypertensive, and a heavy smoker came to the emergency room (ER) complaining of progressive, severe compressing retrosternal chest pain and shortness of breath. What is the first investigation to be ordered for this patient?

- a. Serum troponin